Rest Framework

- What is an API (Application Programming Interface)?
  - ✓ An API is a way for different software programs to talk to each other. It allows one app to request data or services from another. For example, a weather app uses an API to get weather data.

- Types of APIs: REST, SOAP.
  - ✓ REST APIs use simple HTTP methods like GET and POST and are easy to use. SOAP APIs are more strict and use XML format with higher security. REST is more popular today because it is faster and simpler

- Why are APIs important in web development?
  - ✓ APIs help connect frontend and backend systems. They allow apps to share data and use external services. Without APIs, modern web apps would not work smoothly

- Understanding project requirements.
  - ✓ This means clearly knowing what the project should do. It includes features, user needs, and goals. Good understanding helps avoid mistakes later

- Setting up the environment and installing necessary packages.
  - ✓ You install tools like Python, Django, and libraries needed for the project. This creates a working space for development. It ensures your code runs properly

- What is Serialization?
  - ✓ Serialization means converting data into a format like JSON. This makes it easy to send data over the internet. It is commonly used in APIs

- Converting Django QuerySets to JSON.
  - ✓ Django QuerySets store database data. We convert them into JSON so they can be sent through APIs. This is usually done using serializers

- Using serializers in Django REST Framework (DRF)
  - ✓ Serializers convert complex data into JSON and vice versa. They also validate input data. DRF serializers make API development easier

- Sending and receiving responses in DRF.
  - ✓ DRF uses request and response objects. It sends data in JSON format to the client. It also receives user input from API requests

- Understanding views in DRF: Function-based views vs Class-based views
  - ✓ Function-based views are simple and easy to write. Class-based views are more organized and reusable. Both are used to handle API requests

- Defining URLs and linking them to views
  - ✓ URLs are endpoints where users access your API. Each URL is connected to a view. This tells Django what code to run when a request comes.

- Adding pagination to APIs to handle large data sets.
  - ✓ Pagination splits large data into smaller pages. It improves performance and loading speed. Users can navigate data easily

- Configuring Django settings for database,static files, and API keys.
  - ✓ Settings file controls database connection and project configuration. Static files include CSS and images. API keys are used to connect external services securely

- Setting up a Django REST Framework project.
  - ✓ You install DRF and add it to your Django project. Then create apps, models, and serializers. This prepares your project for building APIs

- Implementing social authentication (e.g., Google, Facebook) in Django.
  - ✓ This allows users to log in using their social accounts. It improves user experience and saves time. Libraries like Django-allauth help implement it

- Sending emails and OTPs using third-party APIslike Twilio, SendGrid
  - ✓ These services help send emails or SMS messages. OTPs are used for verification. APIs make sending messages easy and fast

- REST principles: statelessness, resource-based URLs, and using HTTP methods for CRUD operations.
  - ✓ REST APIs do not store user data between requests (stateless). They use URLs to represent resources. HTTP methods like GET, POST, PUT, DELETE handle actions

- What is CRUD, and why is it fundamental to backend development?
  - ✓ CRUD means Create, Read, Update, Delete. These are basic operations for handling data. Every backend system uses CRUD.

- Difference between authentication and authorization.
  - ✓ Authentication checks who the user is (login). Authorization checks what the user can do (permissions). Both are important for security

- Implementing authentication using Django REST Framework's token-based system.
  - ✓ Users get a token after login. This token is sent with every request. It helps verify the user without logging in again

- Introduction to OpenWeatherMap API and how to retrieve weather data
  - ✓ It provides weather data like temperature and forecasts. You send a request with a city name. It returns weather information in JSON

- Using email sending APIs like SendGrid and Mailchimp to send transactional emails
  - ✓ These APIs help send emails automatically. They are used for notifications and marketing emails. They are easy to integrate

- Introduction to Twilio API forsending SMS and OTPs.
  - ✓ Twilio allows sending SMS and OTPs. It is widely used for mobile verification. It works using simple API calls

- Introduction to integrating payment gateways like PayPal and Stripe.
  - ✓ These services allow online payments. They handle transactions securely. APIs help integrate payments into apps

- Using Google Maps API to display maps and calculate distances between locations.
  - ✓ It helps display maps and locations in apps. You can calculate distance and directions. It is useful for location-based services